

W02 Assignment: Explain Abstraction

What is abstraction and why is it important?

Abstraction is the foundation of object-oriented programming. This programming approach allows us to use an object without worrying about its internal workings. It involves showing only the essentials of an object and hiding the internal details of its implementation. That means, we only need to know what it can do, not how it does it. Just as we use a wireless remote control without knowing how our PlayStation 5 works internally, programming uses classes and methods that do the work for us, hiding the complicated details.

It's important because one of the greatest benefits of abstraction is that it makes programming simpler and more organized, making our lives easier. Instead of thinking about all the details at once, we can work with simple ideas. This makes the code easier to read and understand, and also easier to modify in the future. If something is changed internally, the rest of the program continues to function the same. So the programs are more resilient to errors and can grow without becoming overly complicated or chaotic.

A common use of abstraction is when a program needs to interact with objects without knowing their internal details. Analyzing and studying the topic, this example helped me better understand what abstraction is. In our journaling program, the rest of the system doesn't need to know how the entries are stored internally; it only needs a method to add them.

To give an example of abstraction in C# code, in the journal program, abstraction is applied by encapsulating the list of entries within the "Journal" class and exposing only one public method to interact with it.

```
public class Journal
{
    private List<Entry> _entries = new List<Entry>();

    public void AddEntry(Entry newEntry)
    {
        _entries.Add(newEntry);
    }
}
```

Here, the rest of the program uses:

```
theJournal.AddEntry(newEntry);
and does not directly access _entries.
```

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